

13/10/25

Task 10:- CRUD operations in document databases:-

Aim:- To CRUD operations in document databases.

College Management Database in SQL.

Step 1:- Create Database:-

```
CREATE DATABASE CollegeDB;
```

```
USE CollegeDB;
```

Step 2:- Create Tables.

```
CREATE TABLE students (
```

```
    student-id INT PRIMARY KEY AUTO-INCREMENT,
```

```
    name VARCHAR (100),
```

```
    roll-no VARCHAR (20) UNIQUE,
```

```
    department VARCHAR (50),
```

```
    Age INT
```

```
);
```

```
CREATE TABLE courses (
```

```
    course-id INT PRIMARY KEY AUTO-INCREMENT,
```

```
    course-name VARCHAR (100),
```

```
    course-code VARCHAR (20) UNIQUE,
```

```
    credits INT,
```

```
);
```

```
CREATE TABLE staff (
```

```
    staff-id INT PRIMARY KEY AUTO-
```

```
    INCREMENT,
```

```
    name VARCHAR (100),
```

```
    DOB VARCHAR (50),
```

```
    department VARCHAR (50)
```

```
);
```


Step 3:- Insert Records.

INSERT INTO students (name, roll-no, department, age)
VALUES ('Alice Johnson', 'S101', 'computer science', 20);

INSERT INTO students (name, roll-no, department, age).

('Bob Smith', 'S102', 'Mechanical', 21),

('Charlie Brown', 'S103', 'Civil', 19);

Courses.

INSERT INTO sources (course-name, course-code, credit).

VALUES.

('Database, systems', 'CS301', 4)

('Thermodynamics', 'ME201', 3),

('Structural Analysis', 'CE101', 4);

Staff:-

INSERT INTO staff (name, role, department)

VALUES.

('Dr. Smith', 'professor', 'computer science');

('Dr. John', 'Lecturer', 'Mechanical'),

('Dr. Anne', 'HOD', 'Civil');

Step 4:- Read / Query Records

Select * from students, where department
computer science);

S.no	Name	Roll No	Department	age
1.	Alice Johnson	5101	computer science	20

select * from courses where credits = 9

S.no	course name	course code	credits
1	Database systems	CS 301	4
2	Structural Analysis.	CE 101	11

Step 5:- Update Records

UPDATE students

SET age = 21

WHERE roll no = '5101';

UPDATE courses

SET credits = 4.

WHERE course code = 'ME 201';

Step 6:- Delete Records:-

DELETE FROM students WHERE roll no = '5103';

DELETE FROM courses WHERE course code = 'CE 101'.

DELETE FROM staff WHERE name = 'Dr. John';

VELTECH	
EX No.	105
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	5
RECORD (5)	5
TOTAL (20)	10
SIGN WITH DATE	13/10/18

Result - Thus, the CRUD operations in document Databases is executed successfully.